Loan Approval Risk Analysis

1. Project Goal

The goal of this project is to analyze loan application data to understand the factors that influence loan approval decisions and credit risk.

Python was used for data cleaning and preprocessing, SQL was used to answer key business questions, and Tableau was used to build dashboards showing approval rates, credit score impact, income patterns, and risk indicators.

2. Dataset Summary

- **Row:** 5000
- **Columns:** 11
- **Key Features:**

Demographic Features:

Age

Gender

Education

City

YearsExperience

Financial Features:

Income

LoanAmount

CreditScore

YearsExperience

Loan Features:

LoanAmount
Loan_to_income_ratio

3. Exploratory Data Analysis with Python

Start with data preparation and cleaning in Python

- **Import data using Pandas**
- **Initial Exploration:** used .describe() to get the summary statistics and df.info()

.describe()

	Age	Income	LoanAmount	CreditScore	YearsExperience	LoanApproved
count	5000.000000	4804.000000	5000.000000	4806.000000	5000.000000	5000.000000
mean	43.584600	49738.123022	19870.768600	575.494590	19.599000	0.230200
std	14.919094	15101.361851	8046.542413	160.550839	11.516837	0.421003
min	18.000000	-3731.000000	-10059.000000	300.000000	0.000000	0.000000
25%	31.000000	39608.500000	14455.250000	433.000000	10.000000	0.000000
50%	43.000000	49488.000000	19842.500000	579.000000	20.000000	0.000000
75%	56.000000	59917.000000	25326.750000	712.000000	29.000000	0.000000
max	69.000000	99146.000000	48353.000000	849.000000	39.000000	1.000000

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 10 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   Age              5000 non-null   int64
 1   Income           4804 non-null   float64
 2   LoanAmount       5000 non-null   int64
 3   CreditScore      4806 non-null   float64
 4   YearsExperience  5000 non-null   int64
 5   Gender           5000 non-null   object
 6   Education        4802 non-null   object
 7   City             5000 non-null   object
 8   EmploymentType   5000 non-null   object
 9   LoanApproved     5000 non-null   int64
dtypes: float64(2), int64(4), object(4)
memory usage: 390.8+ KB
```

Checked for Missing data: There was no missing data in the data

df.isnull().sum()

	0
Age	0
Income	196
LoanAmount	0
CreditScore	194
YearsExperience	0
Gender	0
Education	198
City	0
EmploymentType	0
LoanApproved	0

dtype: int64

Income missing value:

Column Standarization: renamed the columns name for better readability

```
# rename columns
df = df.rename(columns={
    'LoanAmount': 'loan_amount',
    'CreditScore': 'credit_score',
    'CoapplicantIncome': 'coapplicant_income',
    'YearsExperience': 'years_of_experience',
    'EmploymentType' : 'employment_type',
    'LoanApproved' : 'loan_approved',
    'Gender' : 'gender',
    'Married' : 'married',
    'Education' : 'education',
    'Age' : 'age',
    'City' : 'city',
    'Income' : 'income'
})
df
```

New column:

- Created **loan_approved_** using order_data(this columns is Yes if the loan is approved and No if the loan is not approved)

4. Data Analysis Using SQL

Credit score vs approval (core risk insight)

```
SELECT COUNT(*) AS total_applicants, loan_approved_ AS approval_status,
ROUND(AVG(credit_score),2) AS avg_credit_score
FROM loan_data_cleaned
WHERE credit_score IS NOT NULL
group by loan_approved_ ;
```

total_applicants	approval_status	avg_credit_score
3849	No	533.72
1151	Yes	708.76

Approval rate by credit score ranges:

```
SELECT
  CASE
    WHEN credit_score >= 800 THEN '800+ (Exceptional)'
    WHEN credit_score >= 740 THEN '740-799 (Very Good)'
    WHEN credit_score >= 670 THEN '670-739 (Good)'
    WHEN credit_score >= 580 THEN '580-669 (Fair)'
    ELSE 'Below 580 (Poor)'
  END AS credit_range,
  COUNT(*) AS total_applicants,
```

```
    ROUND(100.0 * SUM(loan_approved) / COUNT(*), 2) AS approval_rate_pct
FROM loan_data_cleaned
WHERE credit_score IS NOT NULL
GROUP BY 1
ORDER BY MIN(credit_score) DESC;
```

credit_range	total_applicants	approval_rate_pct
800+ (Exceptional)	456	48.90
740-799 (Very Good)	519	48.75
670-739 (Good)	621	49.76
580-669 (Fair)	806	37.47
Below 580 (Poor)	2598	2.46

Loan-to-income ratio vs approval:

```
SELECT
    CASE
        WHEN (loan_amount / income) <= .50 THEN 'Low Risk'
        WHEN (loan_amount / income) <= .75 THEN 'Moderate Risk '
        WHEN (loan_amount / income) <= .85 THEN 'High Risk'
        ELSE 'Extreme Risk'
    END AS lti_category,
    COUNT(*) AS total_applicants,
    ROUND(100.0 * SUM(loan_approved) / COUNT(*), 2) AS approval_rate_pct
FROM loan_data_cleaned
WHERE income > 0
GROUP BY 1
```

ORDER BY MIN(loan_amount / income) ASC;

lti_category	total_applicants	approval_rate_pct
Low Risk	3348	26.40
Moderate Risk	1163	19.52
High Risk	175	10.86
Extreme Risk	311	6.75

Approval rate by employment type

```
SELECT
    employment_type,
    COUNT(*) AS total_applicants,
    SUM(loan_approved) AS approved_count,
    ROUND(100.0 * SUM(loan_approved) / COUNT(*), 2) AS approval_rate_percent
FROM loan_data_cleaned
GROUP BY employment_type
ORDER BY approval_rate_percent DESC;
```

employment_type	total_applicants	approved_count	approval_rate_percent
Salaried	1610	536	33.29
Self-Employed	1730	563	32.54
Unemployed	1660	52	3.13

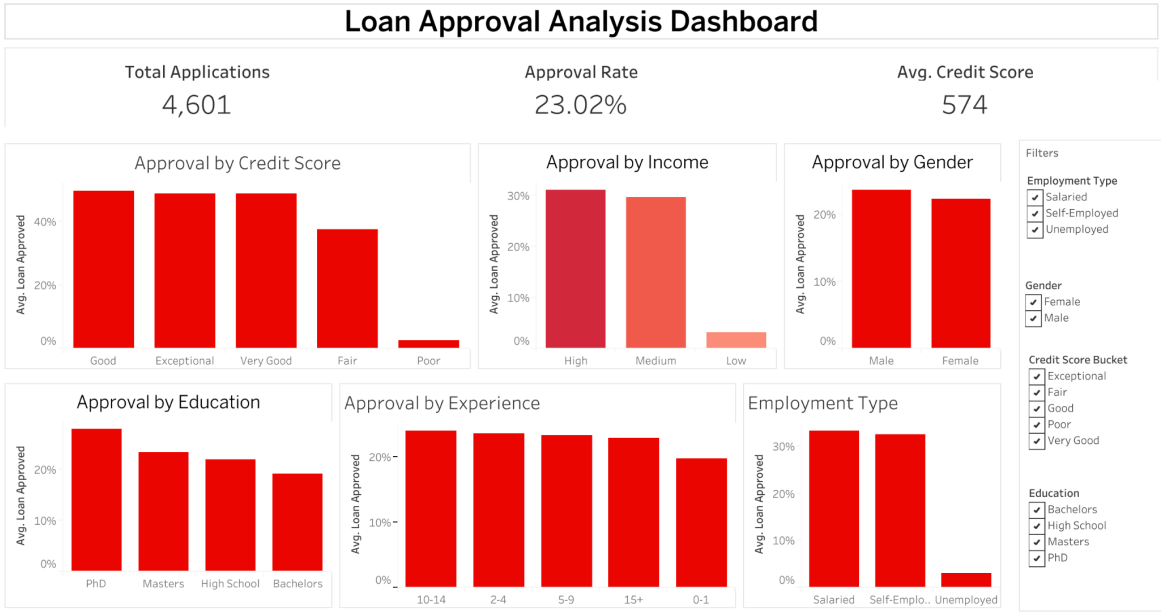
City level loan to income + approval rate

```
SELECT
    city,
```

```
    COUNT(*) AS total_applicants,
    ROUND(AVG(loan_amount / income), 2) AS avg_lti_ratio,
    ROUND(100.0 * SUM(loan_approved) / COUNT(*), 2) AS approval_rate_percent
FROM loan_data_cleaned
WHERE income > 0 -- Avoid division by zero errors
GROUP BY city
ORDER BY approval_rate_percent DESC;
```

city	total_applicants	avg_lti_ratio	approval_rate_percent
San Francisco	1256	0.62	23.89
Chicago	1303	0.45	23.33
New York	1206	0.45	23.05
Houston	1232	0.45	21.83

5. Dashboard in Tableau



6. Business Recommendations

The analysis shows that credit score, income, employment type, and loan-to-income ratio are the strongest predictors of loan approval. By applying minimum credit score rules, limiting loan size based on income, and requiring stable employment, the bank can reduce financial risk and improve loan approval decisions.